

1. $M_n(\mathbb{Z}) \rightarrow$ set of all $n \times n$ matrices with integer entries is also a non-commutative ring with unity.

2. $A = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix} \mid x \in \mathbb{R} \right\}$ is a commutative ring with unity.

$(A, +)$ is an abelian group.

Let $X, Y \in A$ s.t. $X = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$, $Y = \begin{bmatrix} y & y \\ y & y \end{bmatrix}$

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} y & y \\ y & y \end{bmatrix} = \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} = \begin{bmatrix} y & y \\ y & y \end{bmatrix} \begin{bmatrix} x & x \\ x & x \end{bmatrix}$$

Let E be identity of A , s.t. $E = \begin{bmatrix} e & e \\ e & e \end{bmatrix}$

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} e & e \\ e & e \end{bmatrix} = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$$

$$\begin{bmatrix} 2xe & 2xe \\ 2xe & 2xe \end{bmatrix} = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$$

$$2xe = x \implies e = \frac{1}{2}$$

$$\therefore E = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

For making $(A, \cdot)$ abelian group:

If $Y = X^{-1}$ then

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} y & y \\ y & y \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$2xy = \frac{1}{2} \implies y = \frac{1}{4x} \quad (x \neq 0)$$

$$\therefore Y = \begin{bmatrix} \frac{1}{4x} & \frac{1}{4x} \\ \frac{1}{4x} & \frac{1}{4x} \end{bmatrix}$$

8) $A = \left\{ \begin{bmatrix} x & 0 \\ 0 & 0 \end{bmatrix} \mid x \in \mathbb{R} \right\}$ is a commutative ring with unity.

Here $E = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

9) $R = \{0\}$ is a ring with unity 0, called **trivial ring** or **zero ring**.

If $R \neq \{0\}$, then R is called **non-trivial ring**.

10) $\mathbb{Z}[x]$ = set of all polynomials with integer coefficients.

$(\mathbb{Z}[x], +, \cdot)$ is a commutative ring with unity 1.

11) $\mathbb{R}[x], \mathbb{Q}[x], \mathbb{C}[x]$ are also commutative rings with unity.

12) $C[a, b]$ = set of all continuous real-valued functions defined on $[a, b]$.

Define:

$$(f + g)(x) = f(x) + g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x) \quad \forall x \in [a, b]$$

Then $(C[a, b], +, \cdot)$ is a commutative ring with unity $e(x) = 1, \forall x \in [a, b]$.

Properties of a Ring:

Let $(R, +, \cdot)$ be a ring. Then $\forall a, b, c \in R$:

1. $a \cdot 0 = 0 \cdot a = 0$

2. $a(-b) = (-a)b = -(ab)$

3. $(-a)(-b) = ab$

4. $a(b - c) = ab - ac$

5. $(b - c)a = ba - ca$

Proof of Properties:

1. $a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$
 $\Rightarrow a \cdot 0 + 0 = a \cdot 0 + a \cdot 0$
 $\Rightarrow a \cdot 0 = 0$ (by left cancellation in $(R, +)$).
Similarly, $0 \cdot a = 0$.
2. $a \cdot (-b) + ab = a(-b + b) = a \cdot 0 = 0$
 $\Rightarrow a(-b) = -(ab)$.
Similarly, $(-a)b = -(ab)$.
3. $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.
4. $a(b - c) = a(b + (-c)) = ab + a(-c) = ab - ac$.
5. $(b - c)a = (b + (-c))a = ba + (-c)a = ba - ca$.

Zero Divisors:

Let R be a ring. A non-zero element $a \in R$ is said to be a **zero divisor** (or divisor of zero) if there exists a non-zero element $b \in R$ such that:

$$ab = 0 \quad \text{or} \quad ba = 0$$

- If $ab = 0$ with $a \neq 0, b \neq 0$, then a is called a **left zero divisor** and b is called a **right zero divisor**.

Examples:

1. In $(\mathbb{Z}_6, +_6, \times_6)$, elements 2, 3, 4 are zero divisors:

$$2 \times_6 3 = 0, \quad 3 \times_6 4 = 0$$

2. In $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, there are no zero divisors.

Integral Domain:

A commutative ring R with unity ($1 \neq 0$) is called an **integral domain** if it has no zero divisors.
i.e., $\forall a, b \in R$,

$$ab = 0 \implies a = 0 \quad \text{or} \quad b = 0$$

Examples of Integral Domains:

1. $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ are integral domains.
2. $(\mathbb{Z}_p, +_p, \times_p)$ is an integral domain if and only if p is a prime.
3. $\mathbb{Z}[x]$, $\mathbb{Q}[x]$, $\mathbb{R}[x]$ are integral domains.
4. $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ (Gaussian integers) is an integral domain.

Non-examples:

1. $(\mathbb{Z}_n, +_n, \times_n)$ where n is composite is **not** an integral domain.
2. $M_n(\mathbb{R})$ ($n \geq 2$) is not an integral domain (neither commutative nor free of zero divisors).

Theorem:

A commutative ring R is an integral domain if and only if the cancellation law holds in R :

$$ab = ac \quad (a \neq 0) \implies b = c$$

Proof:

($\implies$) Let R be an integral domain and let $ab = ac$ with $a \neq 0$.

$$\begin{aligned} ab - ac = 0 &\implies a(b - c) = 0 \\ &\implies b - c = 0 \quad (\text{since } a \neq 0 \text{ and } R \text{ has no zero divisors}) \\ &\implies b = c \end{aligned}$$

($\impliedby$) Conversely, suppose the cancellation law holds in R .

Let $ab = 0$ with $a \neq 0$.

Then $ab = a \cdot 0$.

Since $a \neq 0$, by cancellation law:

$$b = 0$$

Hence, R has no zero divisors, so R is an integral domain.